Heat Exchange Cabinet Coolant Deionization

(Click here for Summary Version)

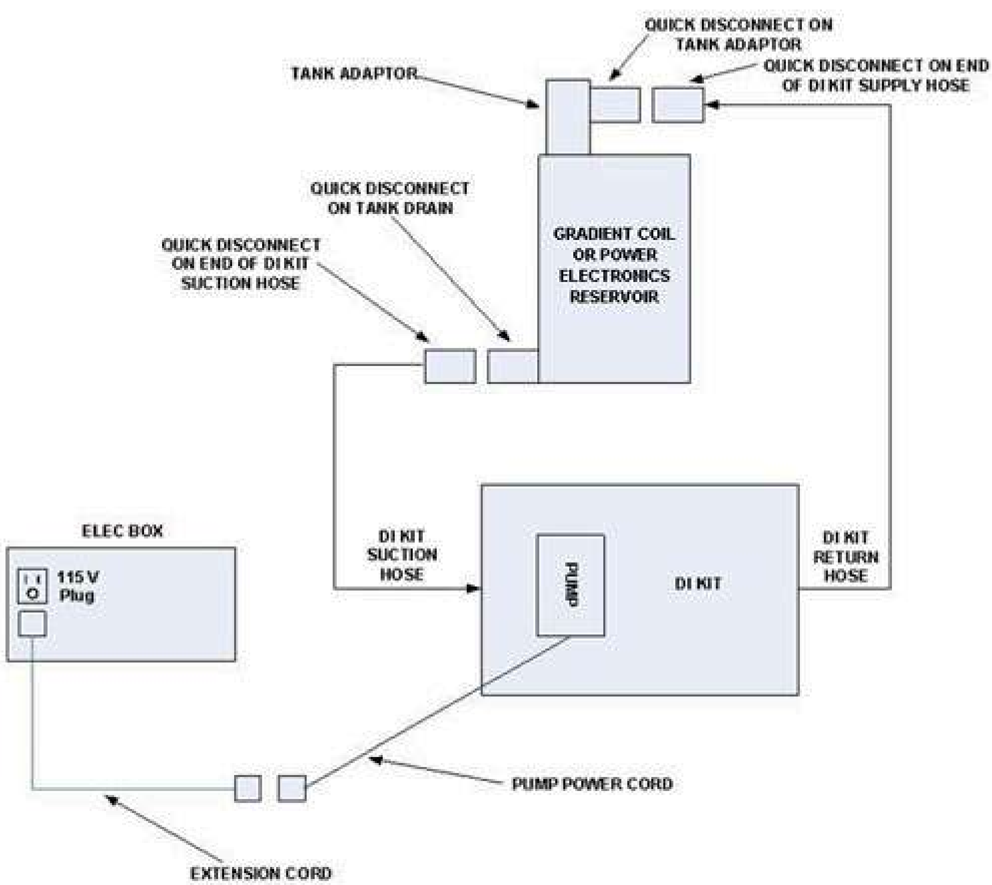
1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	5 mins	90-180 mins, depending on initial resistivity value	10 mins

2 Overview

This procedure describes the process to deionize and filter the coolant for the Gradient Coil (GC) loop and Power Electronics (PE) loop. The Deionization Kit will hook-up to each of the cooling loops to increase the electrical resistivity level and to filter the coolant. Chemical additives will also be added back into each cooling loop. A general schematic of the setup for a cooling loop is shown below. Deionization can be performed while other planned maintenance tasks are being done in parallel, since scanning is still allowed when the tool is hooked up and running.

Illustration 1: General Schematic



3 Preliminary Requirements

3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
DVMR Deionization Kit	1	-	5224090	-

Item	Qty	Effectivity	Part#	Manufacturer
Deionization Filter Cartridge Kit	1	-	5264701	-
5 gallon pail	1	-	2239133	-
Funnel (inside the HEC)	1	-	-	-
Water Removal Pump Kit (inside the HEC)	1	-	-	-

4 Procedure

4.1 Deionization and Filtering the Gradient Coil Coolant

1. Open the Heat Exchange Cabinet (HEC) door. Open the Deionization Kit case and place on the floor in front of the cabinet. Position case so the pump is closest to the floor and horizontal to the floor (see Illustration 2).

Illustration 2: Deionization Kit

2. Uncoil the hoses by unfastening the Velcro straps.
3. Using the wrench tool included with the kit, unscrew both the DI and Filter housings. Each housing may need to be pulled away from the case to fit the wrench behind.
4. Open the package for the Deionization Cartridge, place into the housing. Screw back into the top labeled Deionization. Be sure to tighten fully.

   NOTE: Check that the arrow on the DI cartridge is pointing UP.

5. Open the package for the Filter cartridge, place into the housing and screw back into the top labeled Filter. Be sure to tighten fully.
6. Remove the hex plug from the tank fill port. Thread the DI tank adaptor into the GC tank fill port (see Illustration 3).

   NOTE: Only turn the DI tank adaptor into the fill port two or three revolutions leaving the QD facing out of the HEC cooling cabinet. Ensure that the vent fitting on the tank adapter is open.

Illustration 3: DI Tank Adaptor

7. Connect the DI Kit coolant suction line quick disconnect to the drain port quick disconnect on the side of the GC reservoir. Illustration 4 shows the correct position for the connection, which can go to either tank drain location. Illustration 5 shows where NOT to make the connection.

NOTE: You can connect to the drain port displayed on the left or right (below), depending on the cabinet version.

Illustration 4: Correct Connection for GC Drain Port

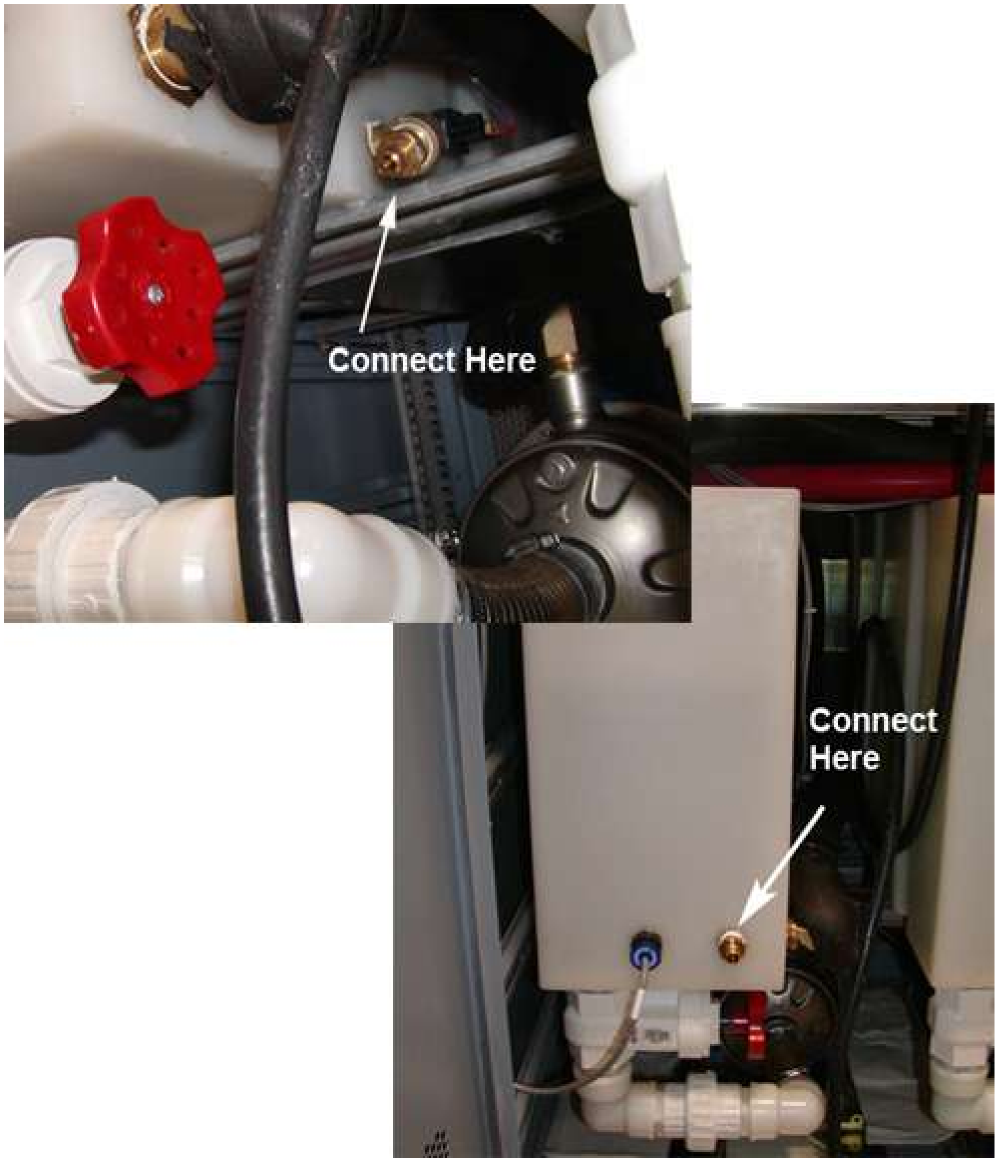


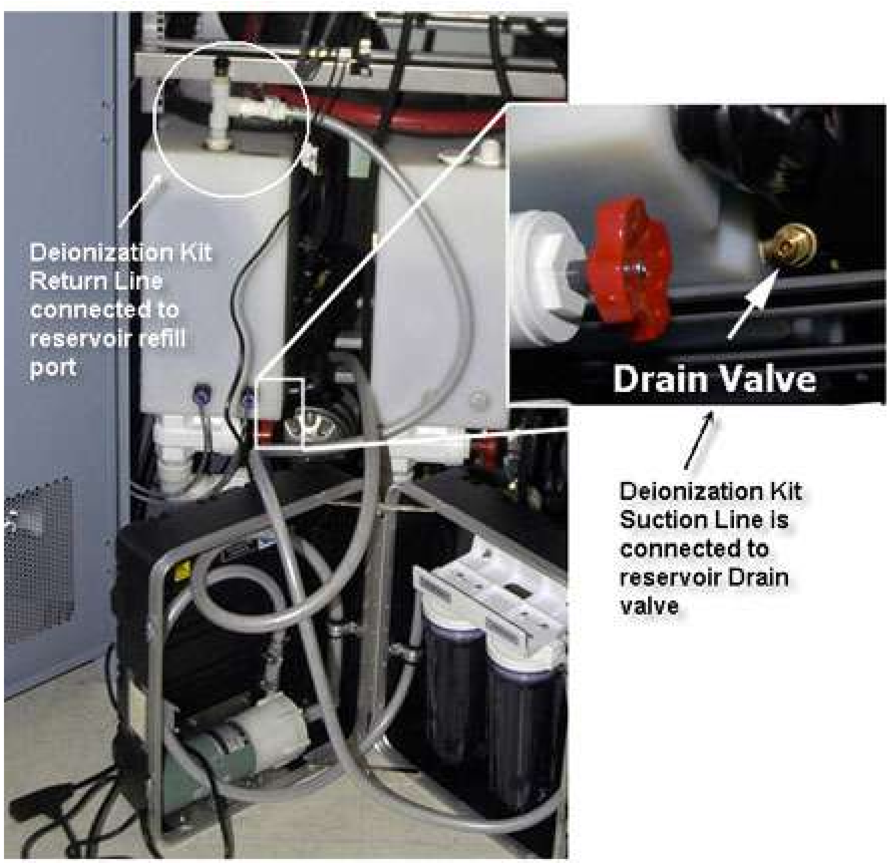
Illustration 5: Incorrect Connection for GC Drain Port

8. Connect the DI Kit coolant return hose quick disconnect to the fill port of the GC reservoir.

9. Observe the suction line. Coolant should be trickling into the suction line. If coolant is observed in the suction line at the pump, the pump is sufficiently primed. If no coolant is observed in the suction line, verify that the return line is connected properly to the reservoir refill port.

   NOTE: DO NOT attempt to run the DI kit pump until the pump is primed.

Illustration 6: Drain Valve



10. Connect power cords:

    a. Connect power cord to extension power cord.

    b. Connect extension cord to DI Kit power outlet located at the top of power box of the HEC.

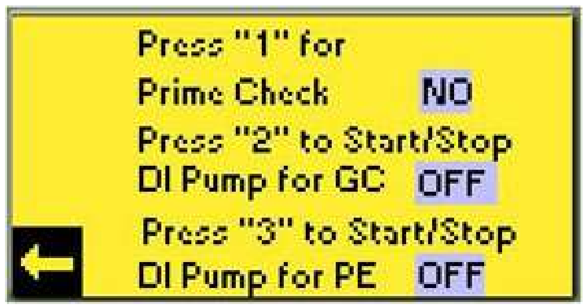
11. Follow these steps on the HEC control screen to start the GC Deionization. (Press <ESC> multiple times if not at the main menu.) Select: <F4 SERVICE>, <F6 - DI Kit>. Press 1 to toggle prime check to **OK**. Press 2 to start GC **ON**.

    NOTE: Only toggle the prime check to **OK** if the pump is primed.

12. The deionization process for the Gradient Coil will run until the resistivity is brought back into specification. This is controlled by PLC and will turn off automatically.

    a. [Illustration 7] shows the DI Kit screen in the service menu on the HEC. The values that say *OFF* will say *ON* when the DI pump is running. These values will return to *OFF* when the pump is off.

    Illustration 7: DI Kit Screen

    b. To check the progress and watch the displayed resistivity value, select the following keys from the main startup screen in this sequence:

        i. **F3** Monitor

        ii. **F2** Gradient Coil

        iii. **F6** Next Screen

NOTE: The deionization process for the Gradient Coil usually takes about 1 hour (but can take up to 3 hours, depending on initial resistivity) and can be done while the system is still in use. The process not only removes coolant contaminants but also coolant additives. Because of this, the procedure will run until the resistivity is > 950kΩ-cm. Re-introducing the additives later will greatly decrease the coolant resistivity to the typical 7kohm cm to 12kohm cm level.

13. Disconnect the quick disconnect (QD) of the DI kit suction line from the QD at the reservoir drain port.

14. Disconnect the QD of the DI kit return line from the tank adaptor at the tank fill port.

15. Remove the tank adaptor from the tank fill port connection.

CAUTION

Hazardous Chemicals
Chemicals may cause injury.
Wear goggles and gloves when pouring chemicals.

16. Place the funnel into the fill port of the GC reservoir. Empty the contents of the Gradient Coil Deionization Chemicals additive into the GC reservoir in small amounts to allow the chemicals to be diluted throughout the system.

    NOTE: If you are scanning, keep an eye on the resistivity value since it will drop rapidly once chemicals are added.

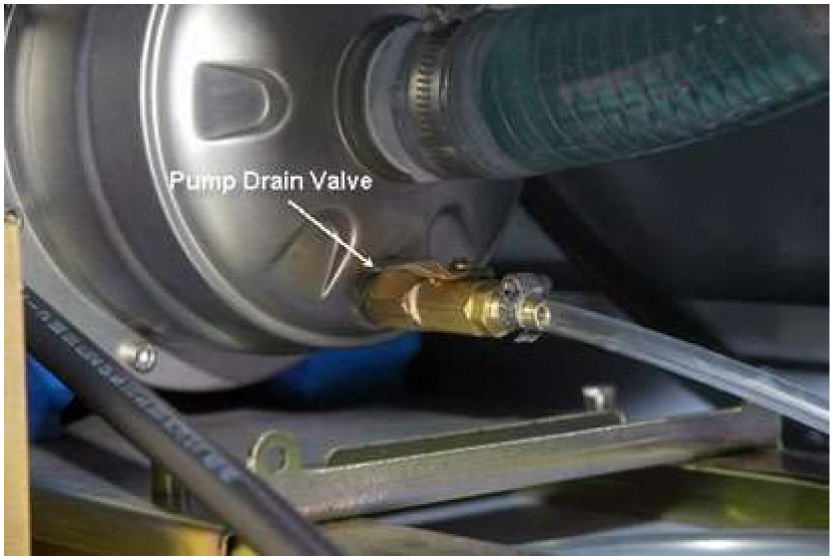
    If the chemical contents are emptied into the reservoir too fast, the resistivity value may drop to a trip level, thus interrupting the customer scan. The resistivity should be in the 7kΩ-cm to 12kΩ-cm range after adding the GC additives.

17. The chemical container must be rinsed three times with the gradient coil water prior to disposal to avoid any environmental hazards:

    a. Open the GC pump drain valve and fill the container.

    b. When the container is nearly full, close the valve and empty the contents of the container into the GC reservoir. Repeat this step two more times. The chemical container may now be disposed.

Illustration 8: Pump Drain Valve



18. Remove the funnel and replace the hex plug for the GC reservoir refill port.

4.2 Deionization and Filtering the Power Electronics Coolant

NOTE: The Power Electronics coolant should likewise be filtered using the deionization kit in order to keep the coolant clean and to refresh the coolant additives.

1. Remove the hex plug from the PE tank fill port. Thread the DI tank adaptor into the tank fill port (see Illustration 9).

    NOTE: Only turn the DI tank adaptor into the fill port two or three revolutions leaving the QD facing out of the HEC cooling cabinet. Ensure that the vent fitting on the adapter is open.

Illustration 9: DI Tank Adaptor

2. Connect the DI Kit coolant suction line quick disconnect to the drain port quick disconnect on the side of the PE reservoir. Illustration 10 shows the correct position for the connection, which can go to either tank drain location. Illustration 11 shows where NOT to make the connection.

NOTE: You can connect to the drain port displayed on the left or right (below), depending on the cabinet version.

Illustration 10: Correct Connection for PE Drain Port

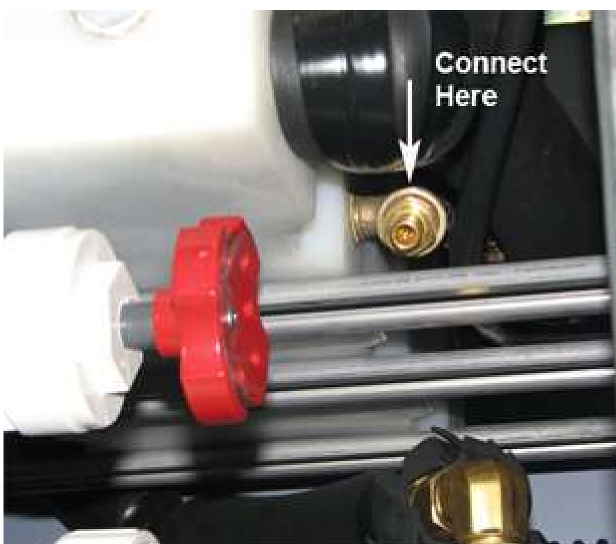


Illustration 11: Incorrect Connection for PE Drain Port

3. Connect the DI Kit water return line quick disconnect to the fill port of the PE reservoir.

4. The Deionization Kit pump should already be primed from completing Deionization and Filtering the Gradient Coil Coolant. Follow these steps on the control screen to start the PE Deionization:

   NOTE: Press <ESC> multiple times if not at the main menu.

   a. Select: <F4 SERVICE>, <F6 - DI Kit>.

   b. Press 1 to toggle prime check to **OK**.

   c. Press 3 to start **PE ON**.

5. Illustration 7 shows the DI Kit screen in the service menu on the HEC. The values that say *OFF* will say *ON* when the DI pump is running. These values will return to *OFF* when the pump is off.

NOTE: The filtering process for the power electronics will run for a set time. The filtering process can be done while the system is in use and will usually take 30 minutes to complete.

6. Disconnect the QD of the DI kit suction line from the QD at the tank drain after deionization is complete.

7. Disconnect the QD of the DI kit discharge line from the tank adaptor at the tank fill port.

8. Remove the tank adaptor from the tank fill port connection.

CAUTION
Hazardous Chemicals
Chemicals may cause injury.
Wear goggles and gloves when pouring chemicals.

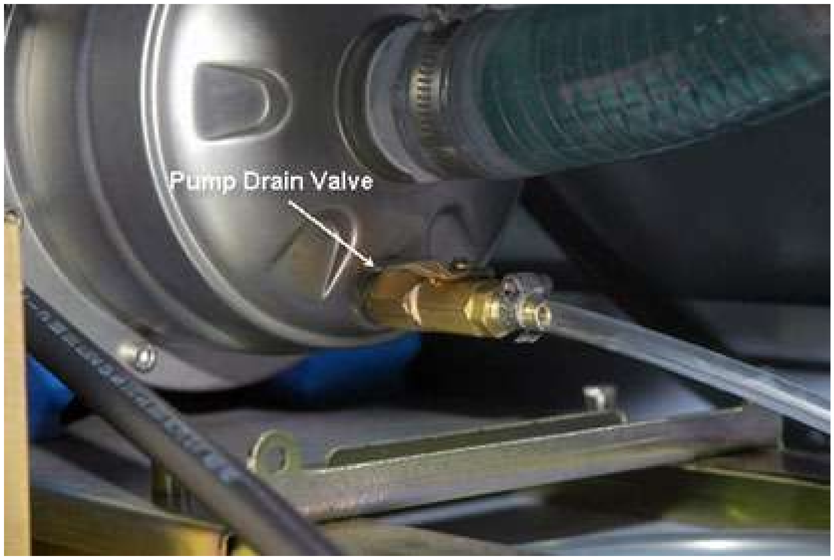
9. Place the funnel into the refill port of the PE reservoir. Empty the contents of the Power Electronics Deionization Chemicals container into the PE reservoir.

10. The chemical container must be rinsed three times prior to disposal to avoid any environmental hazards:

    a. Open the PE pump drain valve and using the hose fill the container.

    b. When the container is nearly full, close the valve and empty the contents of the container into the PE reservoir. Repeat this step two more times. The chemical container may now be disposed.

Illustration 12: Pump Drain Valve



11. Remove the funnel and replace the hex plug for the PE reservoir fill port.

12. Follow the procedure to drain the DI Kit (Section 4.3).

4.3 Draining the DI Kit

1. Unplug the DI Kit power cord and extension cord.

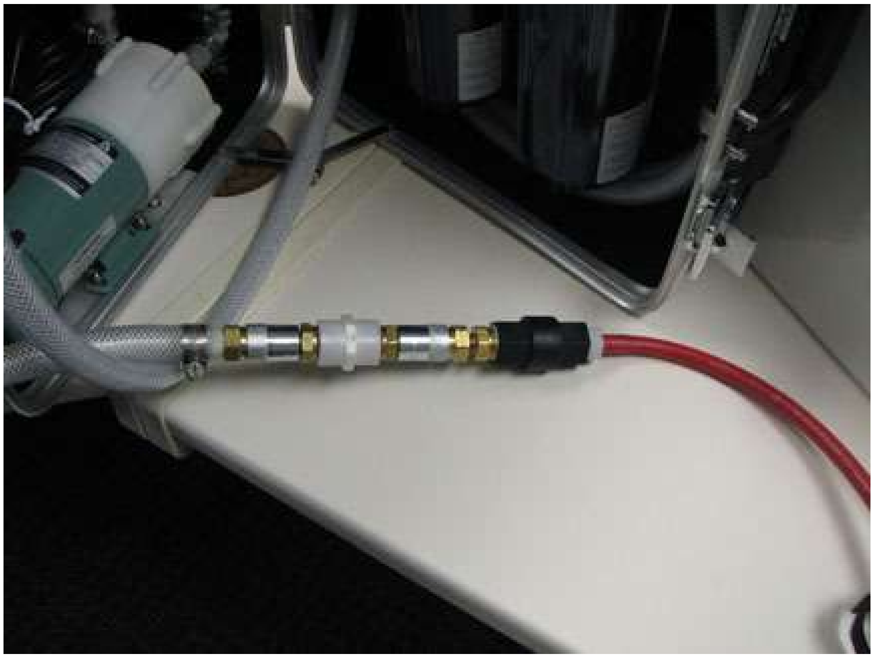
2. Remove the Water Removal Pump Kit from the HEC.

3. Attach the Water Removal Pump quick disconnect adapter to the DI Kit suction line. See Illustration 13.

Illustration 13: Connection of Water Removal Adapter to DI Kit

4. Attach the water removal pump suction line to the hose adapter of the DI Kit suction line via the quick disconnect and place the return line in the bucket to allow coolant to drain. See Illustration 14.

Illustration 14: Connection of Water Removal Pump to Hose Adapter of DI Kit



5. Begin pumping the Water Removal Pump to remove coolant from the kit. Pump until no coolant is observed flowing from the DI Kit return hose.

6. Using the wrench tool, do the following:

   a. Remove the DI housing and filter housing from the kit.

   b. Discard any residual fluid in the housings into the bucket.

   NOTE: This fluid is bio-friendly and can be discarded down a normal sewage drain. Although the water is treated for bateria and water corrosion, the levels of concentrate are safe for water and sewer systems disposal.

   c. Discard the DI filter cartridge and the filter cartridge.

   d. Re-install the empty DI housing and Filter housing so they are hand tight.

7. Disconnect the Water Removal Pump from the adapter for the DI Kit. Disconnect the adapter for the DI Kit from the DI Kit suction line.

5 Finalization

1. Wipe down all wet surfaces.

2. Close the HEC door.

3. Coil the power extension cord and place on top of the DI & Filter support bracket.

4. Coil the DI Kit hoses using the Velcro straps. Close the case and clasps.

Illustration 15: Packing the DI Kit

Back to top